Data Engineering Internship

As a Data Engineering Intern, you will undertake a comprehednsive project aimed at building a robust data pipeline for Formula 1 data. The project encompasses multiple stages, including data modeling, ETL (Extract, Transform, Load) process creation using Apache Airflow and Python with Kafka, and the development of a PowerBI report.

Project Tasks:

Data Modeling:

- Utilize the provided Formula 1 dataset to design and implement a star schema.
- Identify relevant dimensions such as Race, Driver, Team, Circuit, and Time, along with the necessary fact tables.
- Define primary and foreign keys to establish relationships between dimension and fact tables.

ETL Process with Airflow and Kafka:

- Develop an ETL process using Apache Airflow to automate the extraction, transformation, and loading of data from the Formula 1 dataset into the star schema.
- Utilize Python to write custom operators within Airflow to handle data transformation tasks efficiently.
- Implement Kafka as a messaging system to ensure data streaming capabilities within the pipeline, enabling real-time data processing.

API Data Scraping:

- Identify a suitable API that provides additional Formula 1-related data
- Develop a Python script to scrape data from the chosen API.
- Integrate the scraped data into the existing data pipeline, ensuring compatibility with the established star schema.

PowerBI Report Creation:



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- Utilize PowerBI to create an insightful and interactive report/dashboard based on the enriched Formula 1 dataset.
- Design visualizations to showcase key metrics such as race results, driver performance, team standings, and circuit statistics.
- Ensure the report provides actionable insights for stakeholders, including race strategists, team managers, and Formula 1 enthusiasts.

Deliverables:

- Documented star schema design including entity-relationship diagrams (ERDs) and schema definition.
- Python scripts for ETL processes implemented in Apache Airflow.
- Kafka configuration files and Python scripts for data streaming within the pipeline.
- Python script for API data scraping and integration.
- PowerBI report/dashboard showcasing key Formula 1 metrics and insights.

Evaluation Criteria:

- Completeness and correctness of the star schema design.
- Efficiency and reliability of the ETL process implemented with Airflow and Kafka.
- Effectiveness of API data scraping and integration into the pipeline.
- Quality and clarity of the PowerBI report/dashboard.
- Documentation quality, including code comments and explanatory notes.

Note:

This internship assignment is designed to provide practical experience in various aspects of data engineering, including data modeling, ETL processes, API integration, and data visualization. You are encouraged to seek guidance and assistance from mentors and peers throughout the project duration. Good luck, and have fun exploring the exciting world of Formula 1 data!

